Institute of Management Technology
Hyderabad

Risk Analytics

Research Question:

"What are the key determinants of Non-Performing Assets in selected Indian banks?"

Assignment Submission

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When we think about the banking sector, we think about channelling savings into productive investments. But the major challenge facing the banking sector is the rise in Non-Performing Assets (NPAs), which result from loans in which borrowers fail to meet repayment obligations. If the NPA increases, it weakens banks' financial health, reduces profitability, and creates instability in the financial system.

In recent years, rising NPAs have been a major problem in the Indian banking sector. Over the years, several banks experienced a sharp rise in bad loans, especially after the RBI introduced stricter asset-quality recognition measures. This increase in NPAs affected both public and private sector banks, raising concerns about banking efficiency and credit risk management.

Both bank-specific factors and macroeconomic factors may affect the level of NPAs. For this study, I have considered the Capital Adequacy Ratio (CAR), Bank Size (log return of total assets), Return on Assets (ROA), and Credit Growth as the bank-specific variables. And GDP growth, inflation are considered as the macroeconomic variables.

The study analyses the determinants of NPAs in selected Indian banks over the period 2014-2023. The study includes six banks, three from the public and three from the private sector. State Bank of India(SBI), Punjab National Bank, and Bank of Baroda (BOB) are the public sector banks. And ICICI Bank, HDFC Bank, and Axis Bank are private-sector banks.

The primary software used for the study is 'R'. Various models are run in R during the study, including Fixed Effects, Random Effects, Pooled OLS, and the Hausman Test. The study employs a panel data regression approach, which allows analysis of both cross-sectional differences across banks and differences over time.

Research Question

The main question on which the study is based can be stated as follows:

"What are the key determinants of Non-Performing Assets in selected Indian banks?"

Along with the main research question, sub-questions that will be studied are as follows:

• Do bank-specific factors such as capital adequacy, profitability, and bank size influence NPAs?

• Do macroeconomic variables such as GDP growth and inflation affect NPAs?

• Which factors have the strongest impact on NPAs?

Expected Outcome of the Study

At the start of the study, it is expected that both bank-specific and macroeconomic variables affect or influence the level of NPAs of the banks.

For reference, we can say:

• Higher CAR may reduce NPAs because well-capitalized banks tend to manage risk more effectively.

• Higher ROA may be associated with lower NPAs, as profitable banks usually have better asset quality.

• Rapid credit growth may increase NPAs if banks expand lending aggressively without proper risk assessment.

• Economic growth, i.e., GDP growth, may reduce NPAs because borrowers are more capable of repaying loans during strong economic conditions.

• Inflation may have mixed effects depending on economic conditions.

Models studied and the result analysis

Note: The credit growth of the 2014 for each bank has been made mean in R by using the following code: data$CreditGrowth[is.na(data$CreditGrowth)] <- mean(data$CreditGrowth, na.rm = TRUE)

Pooled OLS

This model gives the simplest approach to estimate the relationship between the dependent variable and the explanatory variables, i.e., in this study, NPA is the dependent variable, and the remaining others are independent variables. It assumes that all observations are homogeneous and ignores individual-specific effects across banks for selected time periods. In other words, the pooled model treats the dataset as a simple cross-sectional dataset without considering bank-specific characteristics.

Model result from R:

```
> #Pooled OLS model
> pooled_model <- plm(NPA~CAR + BankSize + ROA + CreditGrowth + GDPGrowth + Inflation,
+                     data = pdata, model = "pooling")
> summary(pooled_model)
Pooling Model

Call:
plm(formula = NPA ~ CAR + BankSize + ROA + CreditGrowth + GDPGrowth +
    Inflation, data = pdata, model = "pooling")

Balanced Panel: n = 6, T = 10, N = 60

Residuals:
      Min.    1st Qu.     Median    3rd Qu.       Max.
-6.8445464 -1.3005167 -0.0082874  1.0818540  6.3215231

Coefficients:
               Estimate Std. Error t-value  Pr(>|t|)
(Intercept)   6.978754   4.082314  1.7095   0.09321 .
CAR           0.095814   0.179110  0.5349   0.59493
BankSize      0.249316   0.266044  0.9371   0.35295
ROA          -4.130920   0.536781 -7.6957 3.434e-10 ***
CreditGrowth  0.054649   0.032988  1.6566   0.10350
GDPGrowth    -0.036895   0.077412 -0.4766   0.63560
Inflation    -0.490150   0.278097 -1.7625   0.08375 .
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Total Sum of Squares:    1100
Residual Sum of Squares: 299.51
R-Squared:      0.72772
Adj. R-Squared: 0.6969
F-statistic: 23.6088 on 6 and 53 DF, p-value: 2.274e-13
```

Screengrab 1: Pooled OLS result from R

Pooled OLS model analysis:

The overall model is statistically significant. The F-statistic is 23.61 with a p-value of 2.27e-13, indicating that combining all explanatory variables has a significant effect on NPAs. The model explains a substantial portion of the variation in NPAs, with an R-squared value of 0.7277, meaning that approximately 72.7% of the variation in NPAs is explained by the independent variables included in the model.

Among the explanatory variables, ROA shows a strong and statistically significant relationship with NPAs. The coefficient of ROA is -4.13, and it is significant at the 1% level ($p < 0.001$). This indicates that an increase in profitability is related to a significant reduction in NPAs. In other words, banks with higher profitability tend to maintain better asset quality and lower levels of non-performing loans.

The Inflation rate shows a negative relationship with NPAs, with a coefficient of -0.49, and is weakly significant at the 10% level. This says that higher inflation may slightly reduce the level of NPAs, although the effect is not strongly significant.

The coefficients of CAR, Bank Size, Credit Growth, and GDP Growth are not statistically significant at conventional levels. It says that within the Pooled OLS framework, the explanatory variables do not show a strong independent effect of NPAs.

However, it is important to note that the pooled OLS model assumes that there are no bank-specific effects, which may not be true in banking studies. Different banks may have unique characteristics such as management efficiency, lending policies, and risk management practices that influence NPAs. Ignoring these unobserved heterogeneities may lead to biased or inconsistent estimates.

Fixed Effects Model

This model is estimated to control for unobserved bank-specific characteristics that may influence NPAs. These characteristics may include differences in management practices, risk policies, or operational structures across banks. By allowing each bank to have its own intercept, the Fixed Effects model eliminates the impact of time-invariant factors that are unique to each bank. This allows examination of how changes in explanatory variables within a bank over time affect NPAs.

Model result from R:

```
>
> #Fixed effects model
> fixed_model <- plm(NPA~CAR + BankSize + ROA + CreditGrowth + GDPGrowth + Inflation,
+                    data = pdata, model = "within")
> summary(fixed_model)
Oneway (individual) effect Within Model

Call:
plm(formula = NPA ~ CAR + BankSize + ROA + CreditGrowth + GDPGrowth +
    Inflation, data = pdata, model = "within")

Balanced Panel: n = 6, T = 10, N = 60

Residuals:
    Min.  1st Qu.   Median  3rd Qu.     Max.
-4.20913 -0.86910  0.12676  1.21322  3.80754

Coefficients:
              Estimate Std. Error t-value  Pr(>|t|)
CAR           0.329231   0.171467  1.9201  0.060799 .
BankSize     -1.929107   0.979772 -1.9689  0.054749 .
ROA          -3.122136   0.533448 -5.8527 4.205e-07 ***
CreditGrowth  0.063700   0.023473  2.7138  0.009212 **
GDPGrowth    -0.080666   0.060452 -1.3344  0.188379
Inflation    -0.820220   0.237749 -3.4499  0.001178 **
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Total Sum of Squares:    421.17
Residual Sum of Squares: 129.61
R-Squared:      0.69227
Adj. R-Squared: 0.62174
F-statistic: 17.9965 on 6 and 48 DF, p-value: 8.3958e-11
```

Screengrab 2: Fixed Effects result from R

Fixed Effects Model analysis:

The overall model is statistically significant. The F-statistic is 17.99 with a p-value of 8.3958e-11, indicating that the explanatory variables jointly have a significant effect on NPAs. The R-squared value is 0.692, suggesting that approximately 69.2% of the variation in NPAs within selected banks over time is explained by the independent variables included in the model.

Among the explanatory variables, ROA has a negative and highly significant impact on NPAs. The coefficient of −3.12, significant at the 1% level, indicates that an increase in bank profitability is associated with a substantial reduction in NPAs. This suggests that more profitable banks tend to have better credit risk management and lower levels of non-performing loans. This relation can be explained by using graph afterwards.

Credit Growth shows a positive and statistically significant relationship with NPAs, with a coefficient of 0.0637, significant at the 1% level. This indicates that higher credit expansion may increase the risk of loan defaults, potentially leading to higher NPAs.

Inflation also has a negative and statistically significant effect on NPAs. The coefficient of −0.82, significant at the 1% level, suggests that higher inflation may reduce the burden of debt repayments, helping borrowers service their loans more easily and reducing NPAs.

The variables CAR and Bank Size show weak significance at the 10% level. CAR has a positive coefficient, suggesting that banks with higher capital adequacy may still experience higher NPAs, possibly due to increased lending capacity. On the other hand, the negative coefficient of Bank Size indicates that larger banks may have relatively lower NPAs due to better diversification and risk management practices. However, GDP Growth does not show a statistically significant effect on NPAs in the Fixed Effects model.

Although the Fixed Effects model accounts for unobserved bank-specific heterogeneity, it assumes that these individual effects are correlated with the explanatory variables. In contrast, the Random Effects model assumes that these individual effects are random and uncorrelated with the regressors. To determine which model is more appropriate for the analysis, the study estimates the Random Effects model and subsequently performs the Hausman test.

Random Effects Model

The Random Effects model assumes that individual bank-specific effects are random and uncorrelated with the explanatory variables. Unlike the Fixed Effects model, it allows the inclusion of time-invariant variables and is more efficient when the assumption of no correlation holds. This model is particularly useful when the variation across entities is assumed to be random rather than fixed.

Model result from R:

```
> # Random effects
> random_model <- plm(NPA~CAR + BankSize + ROA + CreditGrowth + GDPGrowth + Inflation,
+                     data = pdata, model = "random")
> summary(random_model)
Oneway (individual) effect Random Effect Model
   (Swamy-Arora's transformation)

Call:
plm(formula = NPA ~ CAR + BankSize + ROA + CreditGrowth + GDPGrowth +
    Inflation, data = pdata, model = "random")

Balanced Panel: n = 6, T = 10, N = 60

Effects:
                 var std.dev share
idiosyncratic  2.700   1.643 0.195
individual    11.140   3.338 0.805
theta: 0.8462

Residuals:
     Min.  1st Qu.   Median  3rd Qu.     Max.
-4.39681 -1.00437 -0.05868  0.99284  3.93571

Coefficients:
              Estimate Std. Error z-value  Pr(>|z|)
(Intercept)  19.858667   8.962850  2.2157  0.026715 *
CAR           0.276710   0.165809  1.6689  0.095147 .
BankSize     -0.986425   0.750555 -1.3143  0.188759
ROA          -3.198203   0.514392 -6.2174 5.053e-10 ***
CreditGrowth  0.062792   0.023313  2.6934  0.007073 **
GDPGrowth    -0.083136   0.058971 -1.4098  0.158604
Inflation    -0.788338   0.230879 -3.4145  0.000639 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Total Sum of Squares:    437.24
Residual Sum of Squares: 141.64
R-Squared:      0.67606
Adj. R-Squared: 0.63939
Chisq: 110.611 on 6 DF, p-value: < 2.22e-16
```

Screengrab 3: Random Effects result from R

Random Effects Model analysis:

The Chi-square statistic is 110.61 with a p-value less than 0.001, indicating that the explanatory variables jointly have a significant effect on NPAs. The R-squared value is 0.676, which means that approximately 67.6% of the variation in NPAs is explained by the independent variables included in the model.

Among the explanatory variables, ROA has a strong and statistically significant negative relationship with NPAs. The coefficient of −3.20, significant at the 1% level, indicates that higher profitability significantly reduces NPAs. This finding suggests that banks with stronger financial performance tend to maintain better asset quality and more effective credit risk management.

Credit Growth shows a positive and statistically significant relationship with NPAs, with a coefficient of 0.0628, significant at the 1% level. This indicates that rapid credit expansion may increase the risk of loan defaults, leading to a rise in NPAs. Inflation also shows a negative and statistically significant effect on NPAs. The coefficient of −0.79, significant at the 1% level, suggests that higher inflation may reduce the real burden of debt repayment, thereby helping borrowers meet their loan obligations and lowering NPAs.

The CAR shows a positive relationship with NPAs and is weakly significant at the 10% level, suggesting that banks with higher capital levels may engage in greater lending, potentially increasing exposure to credit risk.

However, Bank Size and GDP Growth do not show statistically significant effects on NPAs in the Random Effects model, indicating that these variables do not independently influence NPAs within the sample during the study period. The variance decomposition results indicate that approximately 80.5% of the total variation in the error term is due to differences between banks, while 19.5% is due to idiosyncratic variations within banks over time. This highlights the importance of bank-specific effects in explaining variations in NPAs.

Model Selection Test >>> Although both Fixed Effects and Random Effects models provide valuable insights, it is necessary to determine which model fits to given data. To make this determination, the Hausman test is performed. The Hausman test evaluates whether the individual effects are correlated with the explanatory variables. The result of this test helps identify whether the Fixed Effects model or the Random Effects model provides more consistent and efficient estimates.

Hausman test result and analysis:

Hypotheses of the Hausman test are:

• Null Hypothesis(H_0): The Random Effects model is appropriate (no correlation between individual effects and regressors)

• Alternative Hypothesis(H_1): The Fixed Effects model is appropriate (individual effects are correlated with regressors)

If the p-value is less than 0.05, the null hypothesis is rejected, and the Fixed Effects model is preferred. If the p-value is greater than 0.05, the null hypothesis cannot be rejected and the Random Effects model is considered.

Hausman test result:

data: NPA ~ CAR + BankSize + ROA + CreditGrowth + GDPGrowth + Inflation
chisq = 3.177, df = 6, p-value = 0.7863
alternative hypothesis: one model is inconsistent

Hausman test interpretation:

The Hausman test results show a Chi-square statistic of 3.177 with 6 degrees of freedom and a p-value of 0.7863. Since the p-value is significantly greater than the 5% significance level, the null hypothesis cannot be rejected. This indicates that there is no systematic difference between the coefficients of the Fixed Effects and Random Effects models.

Therefore, the assumption that the individual-specific effects are not correlated with the explanatory variables holds, and the Random Effects model is considered appropriate for this study. As a result, the Random Effects model is selected as the final model for analysing the determinants of Non-Performing Assets (NPAs) in the selected banks.

Although the Random Effects model has been selected for the study as the most appropriate specification, it is essential to verify whether the regression assumptions are satisfied. Therefore, several diagnostic tests are conducted to ensure the reliability and robustness of the model results.

These tests include heteroskedasticity, serial correlation, and robust standard error tests.

Heteroskedasticity test:

Heteroskedasticity occurs when the variance of the error terms is not constant across observations. In a panel data model, this problem can arise due to structural differences among cross-sectional units, such as variations in bank size, lending policies, or operational efficiency. It can lead to biased

standard errors, potentially resulting in incorrect statistical inference about the significance of the explanatory variables. Hypothesis of the tests are as follows:

• Null Hypothesis (H_0): Homoskedasticity exists

• Alternative Hypothesis (H_1): Presence of heteroskedasticity

Result from R:

data: random_model BP = 15.258, df = 6, p-value = 0.01835

Since the p-value is less than the 5% significance level, the null hypothesis of homoskedasticity is rejected. This indicates heteroskedasticity in the model. Therefore, the standard errors of the model were corrected using robust standard errors to obtain reliable statistical inference.

Serial Correlation Test:

It refers to the correlation of error terms across time periods. The presence of serial correlation can lead to inefficient estimates and biased standard errors, which may distort hypothesis testing.

• Null Hypothesis (H_0): No serial correlation

• Alternative Hypothesis (H_1): Serial correlation is present

Serial correlation test result:

Breusch-Godfrey/Wooldridge test for serial correlation in panel models
data: NPA ~ CAR + BankSize + ROA + CreditGrowth + GDPGrowth + Inflation
chisq = 16.723, df = 10, p-value = 0.08071
alternative hypothesis: serial correlation in idiosyncratic errors

Since the p-value is greater than the 5% significance level, the null hypothesis cannot be rejected.

Random effects model with robust standard error:

Since heteroskedasticity was detected in the model, robust standard errors were applied to obtain more reliable coefficient estimates and significance levels. Results with robust standard errors indicate that several variables significantly influence NPAs.

ROA continues to show a strong negative and statistically significant effect on NPAs. The coefficient of −3.198 suggests that higher profitability significantly reduces the level of NPAs. This indicates that more efficient and profitable banks tend to maintain better asset quality. Credit Growth has a positive, statistically significant coefficient of 0.0628, suggesting that rapid credit expansion may increase the risk of loan defaults and lead to higher NPAs.

Inflation also shows a negative and statistically significant relationship with NPAs. The coefficient of −0.788 suggests that higher inflation may reduce the real burden of debt repayment, thereby lowering default rates. The CAR is statistically significant at the 5% level, indicating that banks with higher capital adequacy may also experience higher NPAs, possibly due to increased lending activities. Additionally, GDP Growth becomes statistically significant in the robust model, showing a negative relationship with NPAs. This suggests that stronger economic growth may improve borrowers'

repayment capacity and reduce the level of NPAs. However, Bank Size does not show a statistically significant effect on NPAs in the model.

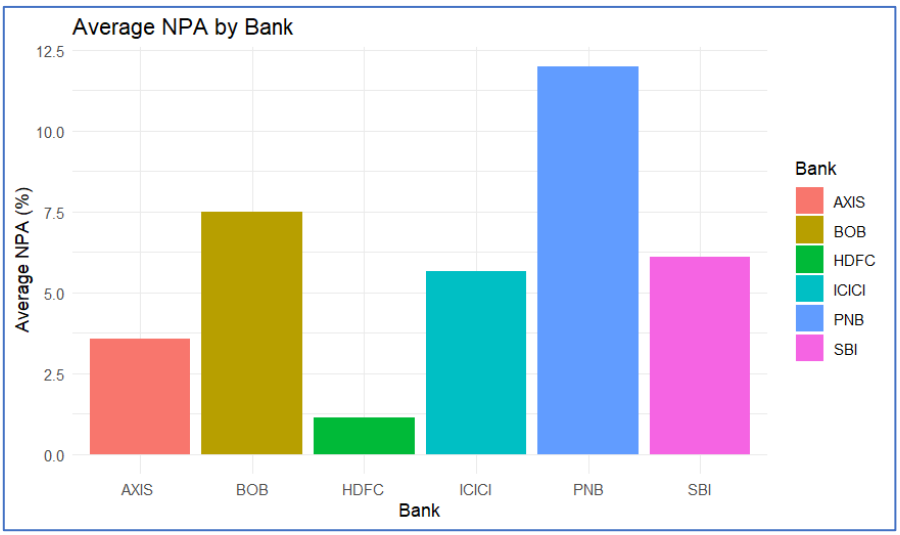
Random effects model result:

```
> #Robust Standard Errors
> coeftest(random_model,vcov. = vcovHC(random_model))

t test of coefficients:

              Estimate Std. Error t value  Pr(>|t|)
(Intercept)  19.858667  11.745340  1.6908  0.096754 .
CAR           0.276710   0.114862  2.4091  0.019501 *
BankSize     -0.986425   0.898370 -1.0980  0.277160
ROA          -3.198203   0.611128 -5.2333 2.906e-06 ***
CreditGrowth  0.062792   0.019180  3.2739  0.001872 **
GDPGrowth    -0.083136   0.036779 -2.2604  0.027928 *
Inflation    -0.788338   0.257973 -3.0559  0.003510 **
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

Screengrab 4: Robust Std Error result from R

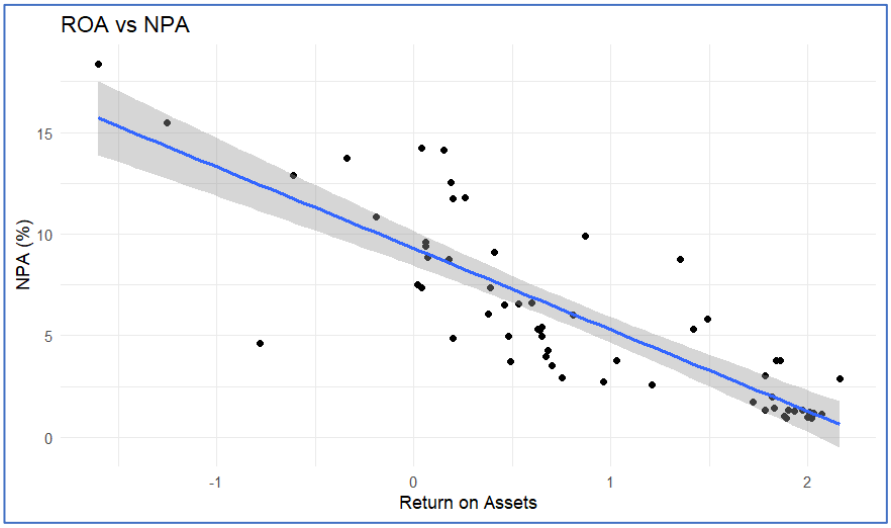
Graphical analysis of NPAs using R

a. Average NPA by banks



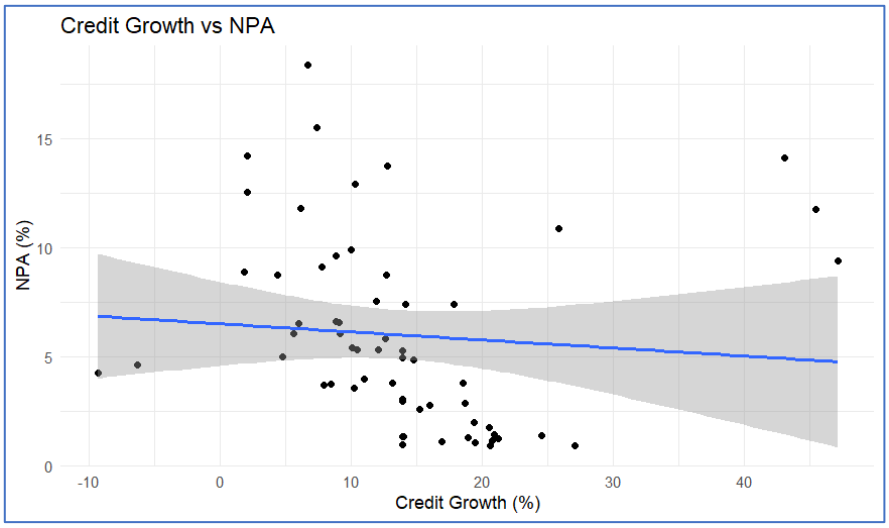
This bar chart illustrates the average level of NPAs for each bank over the study period. We can see that the public sector bank PNB has the highest NPA level, indicating that it experienced the greatest challenges related to loan defaults during the study period. BOB stands in the second-highest place. In contrast, HDFC Bank displays the lowest average NPA level, suggesting stronger credit risk management and more stable asset quality. Overall, the chart indicates that private sector banks (HDFC, ICICI, Axis) tend to exhibit better asset quality than public sector banks (SBI, BOB, PNB) in the sample.

b. ROA and NPA relationship



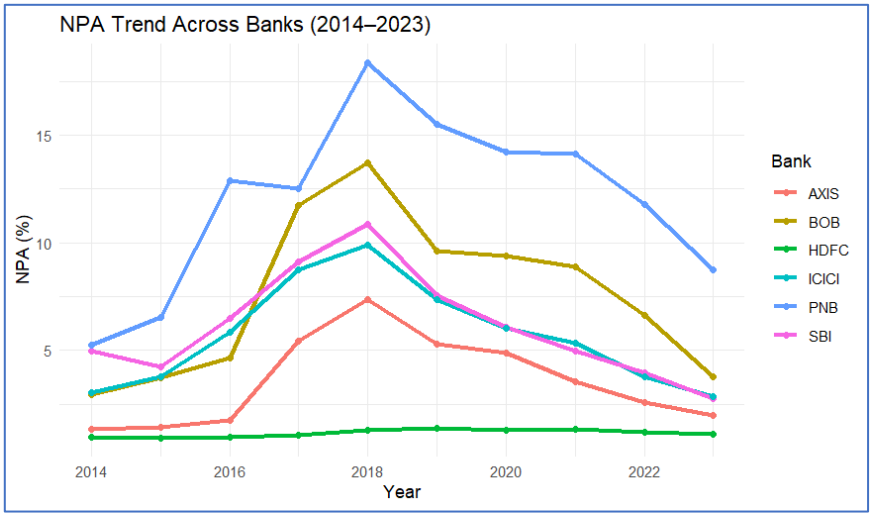
Above scatter plot illustrates the relationship between ROA and the NPA across the banks. The downward slope regression line indicates a clear negative relationship between profitability and NPA levels. This suggests that banks with higher profitability tend to have lower NPAs. In other words, more efficient and profitable banks are generally better at managing credit risk and maintaining the quality of their loan portfolios. This graphical result aligns with the regression results from the econometric analysis, which show that ROA has a statistically significant negative effect on NPAs.

c. Credit growth and NPA relationship



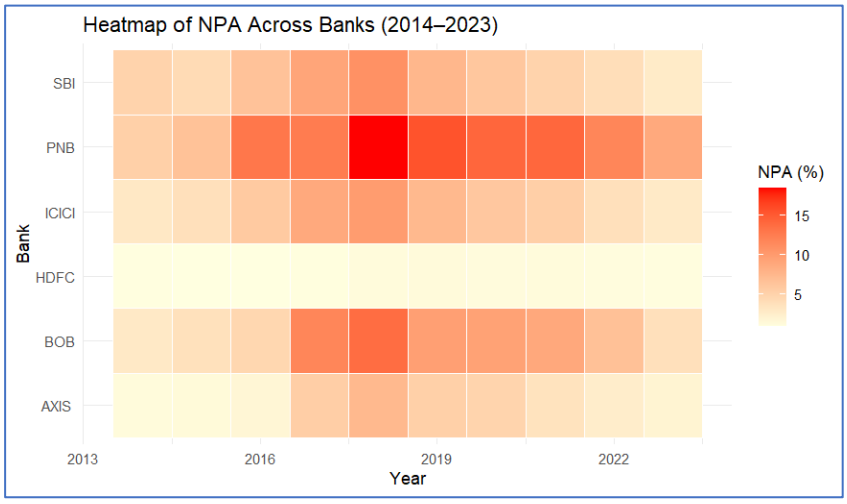
Above scatter plot illustrates relationship between NPA and credit growth. The fitted regression line shows a relatively weak relationship between the two variables, indicating that the impact of credit growth on NPA is not straightforward. While some observations with higher credit growth correspond to moderate or lower NPA levels, there are also instances where increased lending is associated with higher NPAs. This suggests that the effect of credit growth on asset quality may depend on the quality of lending practices rather than the volume of lending alone. Overall, the figure highlights the complex relationship between lending expansion and loan default risk.

d. NPA trend across banks (2014-2023)



The figure shows that NPAs generally increased for most banks between 2015 and 2018, reaching a peak around 2018. This period corresponds to a phase of significant stress in the banking sector, particularly among public sector banks. PNB and BOB exhibit the highest increases in NPAs during this period, indicating greater exposure to loan defaults. After 2018, the NPA levels for most banks begin to decline gradually, suggesting improvements in asset quality, possibly due to regulatory reforms, improved credit monitoring, and recovery mechanisms. HDFC Bank maintains consistently low NPA levels throughout the period, reflecting stronger risk management practices. Overall, the graph demonstrates that while NPAs rose sharply during the mid-period, they gradually declined in the later years across most banks.

e. Heatmap of NPA Across Banks



Above heatmap represents the visual representation of NPA across the time period 2014-23. Darker shades indicate higher levels of NPAs, while lighter shades indicate relatively lower levels. The figure shows significant variation in asset quality across banks during the study period. Public sector banks such as PNB and BOB show noticeably higher NPA levels, particularly around 2017–2019, suggesting a period of increased loan defaults or stress in their loan portfolios. HDFC Bank consistently shows lighter shades, indicating relatively low and stable NPA levels throughout the period. Private sector banks such as ICICI and AXIS show moderate NPA levels with some fluctuations during the middle years of the sample.

Note: R code for the models and graphs is available in the shared R file. After running the above results, they are evaluated.

Conclusion

This particular assignment or study examines the determinants of NPAs using panel data from six banks for the period of ten years. The techniques involved are Pooled OLS, Fixed Effects, Random Effects, and the Hausman test.

The Hausman test results indicate that the Random Effects model is the most suitable specification, suggesting that the individual bank effects are not significantly correlated with the explanatory variables. Hence, the Random Effects model with robust standard errors is used for the final interpretation of results.

Findings reveal that bank profitability, credit growth, inflation, capital adequacy, and economic growth play an important role in determining the level of NPAs.

ROA shows a strong negative relationship with NPAs, which indicates that more profitable banks tend to maintain better asset quality and lower levels of NPAs. Credit Growth shows a positive relationship with NPAs, suggesting that rapid expansion of credit may increase the risk of loan defaults.

Macroeconomic factors also play a role. Inflation and GDP Growth are negatively associated with NPAs, suggesting that favorable macroeconomic conditions can improve borrowers' repayment capacity and reduce default risk. Diagnostic tests indicate heteroskedasticity in the model; therefore, robust standard errors were used to obtain reliable inference. The results remain consistent after correcting for heteroskedasticity, strengthening the validity of the findings.

Overall, the study highlights the importance of both internal bank management factors and external macroeconomic conditions in influencing the level of NPAs in the banking sector.

References:

Inflation (annual %)-India. (2025)

Required link: https://data.worldbank.org/indicator/FP.CPI.TOTL.ZG?locations=IN

GDP Growth (annual %)- India. (2025)

Required link: https://data.worldbank.org/indicator/NY.GDP.MKTP.KD.ZG?locations=IN

Respective annual reports of banks. (2014-2023)